
Examiners' commentaries 2025 (May)

ST2134 Advanced statistics: statistical inference

Important note

This commentary reflects the examination and assessment arrangements for this course in the academic year 2024–25. The format and structure of the examination may change in future years, and any such changes will be publicised on the virtual learning environment (VLE).

Information about the subject guide and the Essential reading references

Unless otherwise stated, all cross-references will be to the latest version of the subject guide (2021). You should always attempt to use the most recent edition of any Essential reading textbook, even if the commentary and/or online reading list and/or subject guide refer to an earlier edition. If different editions of Essential reading are listed, please check the VLE for reading supplements – if none are available, please use the contents list and index of the new edition to find the relevant section.

General remarks

Learning outcomes

At the end of this half course and having completed the essential reading and activities you should be able to:

- explain the principles of data reduction
- judge the quality of estimators
- choose appropriate methods of statistical inference to tackle real problems.

Key steps to improvement

Candidates should pay close attention to the essential topics of sufficient statistics, point estimation, interval estimation and hypothesis testing. Being able to state and prove key theorems and lemmas simply requires close study of these in the subject guide. The point estimation techniques of method of moments estimation and maximum likelihood estimation are standard (and are even introduced in **ST104B Statistics 2** and **ST1215 Introduction to mathematical statistics**), with questions typically varying in terms of the probability distribution from which the random sample is drawn (with many common distributions covered in **ST104B Statistics 2**, **ST1215 Introduction to mathematical statistics** and **ST2133 Advanced statistics: distribution theory**). Pivotal functions feature heavily in Chapter 4 (Interval estimation), and likelihood ratio tests similarly feature heavily in Chapter 5 (Hypothesis testing).

An examination paper attempts to cover as much of the syllabus as possible, but inevitably some parts are left out. As the included parts will vary from year to year it is not enough to practise only

on the past papers. Candidates should practise on all the examples, activities and sample examination questions in the subject guide in order to understand the relevant theory and adequately prepare for the examination. Afterwards you may try one or two past papers for further practice.

The examination paper for this course involves four questions and each of the questions contains several parts which contribute towards the final grade with the different numbers of marks reflecting their difficulty. The difficulty of each part is usually assessed after taking into account the level of thinking related to statistical inference and the extent of algebraic computations required. Take 5–10 minutes to read the whole paper and try to identify the difficulties of each question in comparison to your ability. You will find that some questions could be much easier for you than others.

If you are unable to solve an exercise due to algebraic computations or lack of knowledge of the material of **ST2133 Advanced statistics: distribution theory**, do not abandon an exercise. Write clearly the procedure which should be followed for the relevant statistical inference task(s). Remember that the primary aspect you are being examined on is the ability to *understand* the statistical methodology.

A good knowledge of basic calculus, such as the ability to compute integrals, sums and derivatives, will give you invaluable help for completing the examination paper. Also, the main concepts of **ST2133 Advanced statistics: distribution theory** (random variables and their distribution, expectation, independence, conditional expectation etc.) should be well-understood before attempting this half course. If you are not comfortable with any of these, go back and revise. Memorising basic properties such as the moments and probability mass/density functions of some standard distributions (such as the binomial, Poisson, uniform, normal and gamma) is also a good idea.

When you are trying to prove an equality or inequality be careful of what is on the left-hand and right-hand sides. Justify the steps in the calculations as much as possible. If you cannot get a question right, do not just write in the correct answer without any further explanation. If you just write on the script what you think went wrong you will get more marks. Also, try to avoid writing more than you are asked to write. For example, when asked to *state* a theorem, do not give the proof as well as this simply wastes time.

Be clear in your answers. When giving the probability mass/density function of a random variable you should also state its sample space (also known as its support). When you are deriving an estimator or a sufficient statistic also state the parameter to which it corresponds. Sometimes it is a good idea to underline your final result (if applicable).

Make sure that you are able to provide the definitions of basic concepts such as estimators vs. estimates, confidence intervals and hypothesis tests. Remember that hypotheses are statements about population parameters which are estimated using functions of the data (that is, *statistics*). In the relevant calculations you should always be able to distinguish a statistic from a parameter and be clear on what they represent, respectively.

Practise operations involving products and sums. A substantial number of candidates fail to write them down correctly. Quite often they omit or write the indices incorrectly which inevitably leads to errors. A typical example of this is the following:

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population with probability density function $f_{X_i}(x_i; \theta)$, often written for simplicity as $f_X(x; \theta)$. An observed sample is $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$. The fact that this sample is a random sample indicates that we can write the joint probability density function of $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ as:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_{X_i}(x_i; \theta).$$

Make sure that you write the above and *not* the following:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_X(x; \theta)$$

which can lead to the *incorrect* expression:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = (f_X(x; \theta))^n.$$

Note that if you were asked to provide the log-likelihood function $l_{\mathbf{X}}(\theta; \mathbf{x})$ the calculations would be formed of *sums* rather than *products* (refresh your memory on the properties of logarithmic functions). The correct expression would be:

$$\begin{aligned} l_{\mathbf{X}}(\theta; \mathbf{x}) &= \log(f_{\mathbf{X}}(\theta; \mathbf{x})) \\ &= \log \left(\prod_{i=1}^n f_{X_i}(x_i; \theta) \right) \\ &= \sum_{i=1}^n \log f_{X_i}(x_i; \theta). \end{aligned}$$

Examination revision strategy

Many candidates are disappointed to find that their examination performance is poorer than they expected. This may be due to a number of reasons, but one particular failing is ‘**question spotting**’, that is, confining your examination preparation to a few questions and/or topics which have come up in past papers for the course. This can have serious consequences.

We recognise that candidates might not cover all topics in the syllabus in the same depth, but you need to be aware that examiners are free to set questions on **any aspect** of the syllabus. This means that you need to study enough of the syllabus to enable you to answer the required number of examination questions.

The syllabus can be found in the Course information sheet available on the VLE. You should read the syllabus carefully and ensure that you cover sufficient material in preparation for the examination. Examiners will vary the topics and questions from year to year and may well set questions that have not appeared in past papers. Examination papers may legitimately include questions on any topic in the syllabus. So, although past papers can be helpful during your revision, you cannot assume that topics or specific questions that have come up in past examinations will occur again.

If you rely on a question-spotting strategy, it is likely you will find yourself in difficulties when you sit the examination. We strongly advise you not to adopt this strategy.

Examiners' commentaries 2025 (May)

ST2134 Advanced statistics: statistical inference

Important note

This commentary reflects the examination and assessment arrangements for this course in the academic year 2024–25. The format and structure of the examination may change in future years, and any such changes will be publicised on the virtual learning environment (VLE).

Information about the subject guide and the essential reading references

Unless otherwise stated, all cross-references will be to the latest version of the subject guide (2021). You should always attempt to use the most recent edition of any Essential reading textbook, even if the commentary and/or online reading list and/or subject guide refer to an earlier edition. If different editions of Essential reading are listed, please check the VLE for reading supplements – if none are available, please use the contents list and index of the new edition to find the relevant section.

Comments on specific questions

Candidates should answer all **FOUR** questions: **Question 1** of Section A (40 marks) and all **THREE** questions from Section B (60 marks in total). **Candidates are strongly advised to divide their time accordingly.**

Section A

Answer Question 1 from this section.

Question 1

- (a) Let $X = \{X_1, X_2, \dots, X_n\}$ be a random sample from the continuous uniform distribution on the interval $(-\theta, \theta)$, where $\theta > 0$ is unknown. Determine the method of moments estimator of θ .

(6 marks)

Reading for this question

Section 3.6.5 of the subject guide explains method of moments estimation.

Approaching the question

By symmetry, the first moment of the distribution is 0 for any value of θ , hence we look at the second population moment. This is:

$$E(X^2) = \int_{-\theta}^{\theta} x^2 \frac{1}{2\theta} dx = \left[\frac{x^3}{6\theta} \right]_{-\theta}^{\theta} = \frac{\theta^2}{3}.$$

We estimate this with the second sample moment to obtain:

$$\frac{1}{n} \sum_{i=1}^n X_i^2 = \frac{\hat{\theta}^2}{3} \implies \hat{\theta}^2 = \frac{3}{n} \sum_{i=1}^n X_i^2 \implies \hat{\theta} = \sqrt{\frac{3}{n} \sum_{i=1}^n X_i^2}.$$

Common errors included mistakenly using the first moment (which is zero and unhelpful), incorrectly computing $E(X^2)$, or failing to equate it to the sample second moment. Some omitted the square root when solving for $\hat{\theta}$. Integration errors also occurred.

- (b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_{25}\}$ be a random sample from a $N(\mu, 1)$ distribution. Consider a test of:

$$H_0 : \mu = 8.0 \quad \text{vs.} \quad H_1 : \mu = 9.0$$

that rejects H_0 if the sample mean is greater than 8.5.

- i. Calculate the size of the test, expressing your answer in terms of Φ (i.e. the standard normal cumulative distribution function).

(4 marks)

- ii. Calculate the power of the test, expressing your answer in terms of Φ .

(4 marks)

Reading for this question

Section 5.4 of the subject guide covers relevant material on statistical tests.

Approaching the question

- i. The sampling distribution of $\bar{X}$ is $N(\mu, 1/25)$ or else $N(\mu, 1/5^2)$. Under H_0 this distribution becomes $N(8, 1/5^2)$. The size of the test is equal to $P(\bar{X} > 8.5)$ under H_0 , which is:

$$P(\bar{X} > 8.5 | \mu = 8) = P\left(Z > \frac{8.5 - 8}{1/5}\right) = P(Z > 2.5) = 1 - \Phi(2.5).$$

- ii. Under H_1 , $\bar{X} \sim N(9, 1/25)$ or else $N(9, 1/5^2)$. The power of the test is $P(\bar{X} > 8.5)$ under H_1 , which is:

$$P(\bar{X} > 8.5 | \mu = 9) = P\left(Z > \frac{8.5 - 9}{1/5}\right) = P(Z > -2.5) = 1 - \Phi(-2.5).$$

Common errors included not recognising the sampling distribution of the sample mean, incorrect standardisation when calculating z scores, and mixing up the hypotheses when computing size and power. Some forgot that $1 - \Phi(z) = \Phi(-z)$ by symmetry.

- (c) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with the probability density function:

$$f_X(x; \theta) = \begin{cases} \frac{x}{\theta} e^{-x^2/(2\theta)} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\theta > 0$ is an unknown parameter. Starting with the *unbiased* estimator $\hat{\theta} = X_1^2$, use the Rao–Blackwell theorem to find a better estimator and show that it is consistent.

Hint: You may assume that $\text{Var}(X_i^2) < \infty$ for all $i = 1, 2, \dots, n$.

(12 marks)

Reading for this question

Section 3.5 of the subject guide discusses the Rao–Blackwell theorem.

Approaching the question

We begin by finding a sufficient statistic for θ . Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \prod_{i=1}^n \frac{x_i}{\theta} e^{-x_i^2/(2\theta)} \propto \theta^{-n} e^{-\sum_i x_i^2/(2\theta)}$$

hence by the factorisation theorem $\sum_{i=1}^n X_i^2$ is a sufficient statistic for θ . We now want to take the expectation of the initial estimator conditional on this statistic:

$$\tilde{\theta} = E\left(X_1^2 \mid \sum_{i=1}^n X_i^2\right) = \frac{1}{n} \sum_{j=1}^n E\left(X_j^2 \mid \sum_{i=1}^n X_i^2\right) = \frac{1}{n} E\left(\sum_{j=1}^n X_j^2 \mid \sum_{i=1}^n X_i^2\right) = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

We are told that the original estimator is an unbiased estimator of θ , so $\tilde{\theta}$ is also unbiased since:

$$E(\tilde{\theta}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \frac{1}{n} n E(X_i^2) = \theta \quad \text{because} \quad E(\hat{\theta}) = \theta.$$

We have that the mean squared error of $\tilde{\theta}$ is:

$$\text{MSE}(\tilde{\theta}) = \underbrace{\text{Bias}(\tilde{\theta})}_{=0} + \text{Var}(\tilde{\theta}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i^2) = \frac{1}{n} \text{Var}(X_1^2).$$

This will tend to zero as $n \rightarrow \infty$, since we are told that $\text{Var}(X_i^2)$ is finite. We conclude that $\tilde{\theta}$ is mean-square consistent.

Common errors included not identifying the sufficient statistic using the factorisation theorem, failing to correctly apply the Rao–Blackwell theorem, or incorrectly computing the conditional expectation. Some candidates also confused unbiasedness with consistency or neglected to show mean squared error convergence.

- (d) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the distribution with the probability density function:

$$f_{\mathbf{X}}(x; \theta) = \begin{cases} \theta/(1+x)^{\theta+1} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\theta > 0$ is an unknown parameter. Assuming that n is large, construct an approximate 95% confidence interval for θ .

(14 marks)

Reading for this question

Sections 4.4–4.7 of the subject guide presents relevant material for the construction of interval estimators.

Approaching the question

Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \prod_{i=1}^n \frac{\theta}{(1+x_i)^{\theta+1}} \propto \theta^n \prod_{i=1}^n (1+x_i)^{-\theta}.$$

Hence the log-likelihood function is:

$$l_{\mathbf{X}}(\theta; \mathbf{x}) = \log L_{\mathbf{X}}(\theta; \mathbf{x}) = n \log \theta - \theta \sum_{i=1}^n \log(1+x_i) + c.$$

The score function is:

$$s_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{\partial}{\partial \theta} l_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{n}{\theta} - \sum_{i=1}^n \log(1+x_i).$$

Fisher information is:

$$\mathcal{I}_{\mathbf{X}}(\theta) = -\mathbb{E} \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right) = -\mathbb{E} \left(-\frac{n}{\theta^2} \right) = \frac{n}{\theta^2}.$$

Setting the score equal to zero gives us the maximum likelihood estimator:

$$\hat{\theta} = n \left(\sum_{i=1}^n \log(1 + x_i) \right)^{-1}.$$

The asymptotic sampling distribution of $\hat{\theta}$ is $N(\theta, \mathcal{I}_{\mathbf{X}}(\theta)^{-1}) = N(\theta, \theta^2/n)$, so the quantity:

$$\sqrt{\mathcal{I}_{\mathbf{X}}(\theta)}(\hat{\theta} - \theta) = \sqrt{n} \left(\frac{\hat{\theta}}{\theta} - 1 \right)$$

is approximately $N(0, 1)$ for large n , i.e. it is an approximate pivotal function. We set:

$$P \left(-1.96 < \sqrt{n} \left(\frac{\hat{\theta}}{\theta} - 1 \right) < 1.96 \right) = 0.95$$

and solve the inequalities for θ to obtain the approximate 95% confidence interval:

$$\left[\hat{\theta} \left(1 + \frac{1.96}{\sqrt{n}} \right)^{-1} < \theta < \hat{\theta} \left(1 - \frac{1.96}{\sqrt{n}} \right)^{-1} \right].$$

Common errors included incorrect differentiation when finding the score function, mistakes in computing Fisher information, or using an inappropriate asymptotic distribution. Some candidates failed to correctly rearrange the pivotal quantity or invert the inequalities to construct the confidence interval.

Section B

Answer all three questions from this section.

Question 2

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with probability density function:

$$f_X(x; \mu, \sigma) = \begin{cases} \frac{1}{\sigma} e^{-(x-\mu)/\sigma} & \text{for } x \geq \mu \\ 0 & \text{otherwise} \end{cases}$$

where $\mu \in \mathbb{R}$ is known, and $\sigma > 0$ is an unknown parameter.

- (a) Determine a minimal sufficient statistic for σ .

(5 marks)

- (b) Determine the score function and state whether the maximum likelihood estimator of σ attains the Cramér–Rao lower bound.

Hint: You may use the fact that the maximum likelihood estimator of σ is $\hat{\sigma} = \bar{X} - \mu$.

(7 marks)

- (c) Using the properties of the score function, show that $\mathbb{E}(\bar{X}) = \mu + \sigma$, find the Fisher information, and (without integrating) compute $\text{Var}(\bar{X})$.

(8 marks)

Reading for this question

Section 2.9 of the subject guide covers minimal sufficiency, while Section 3.9 presents maximum likelihood estimation.

Approaching the question

- (a) Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\sigma; \mathbf{x}) = \prod_{i=1}^n \frac{1}{\sigma} e^{-(x_i - \mu)/\sigma} 1_{[\mu, \infty)} \propto \frac{1}{\sigma^n} e^{-n(\bar{x} - \mu)/\sigma}.$$

By the factorisation theorem, $\bar{X}$ is a sufficient statistic for σ and, because it is a scalar, it is minimally sufficient.

- (b) The log-likelihood function is:

$$l_{\mathbf{X}}(\sigma; \mathbf{x}) = \log L_{\mathbf{X}}(\sigma; \mathbf{x}) = -n \log \sigma - \frac{n(\bar{x} - \mu)}{\sigma} + c$$

and hence the score function is:

$$s_{\mathbf{X}}(\sigma; \mathbf{x}) = \frac{\partial}{\partial \sigma} l_{\mathbf{X}}(\sigma; \mathbf{x}) = -\frac{n}{\sigma} + \frac{n(\bar{x} - \mu)}{\sigma^2} = \frac{n}{\sigma^2}(\bar{x} - \mu - \sigma).$$

This is of the form $b(\sigma)(T(\mathbf{x}) - g(\sigma))$, so we conclude that the maximum likelihood estimator of σ (which is given as $\hat{\sigma} = \bar{X} - \mu$) does indeed attain the Cramér–Rao lower bound.

- (c) We know that $E(s_{\mathbf{X}}(\sigma; \mathbf{X})) = 0$, which implies that:

$$E\left(\frac{n}{\sigma^2}(\bar{X} - \mu - \sigma)\right) = 0 \implies E(\bar{X}) = \mu + \sigma.$$

The Fisher information is then:

$$\mathcal{I}_{\mathbf{X}}(\sigma) = -E\left(\frac{\partial}{\partial \sigma} s_{\mathbf{X}}(\sigma; \mathbf{X})\right) = -E\left(\frac{n}{\sigma^2} - \frac{2n(\bar{X} - \mu)}{\sigma^3}\right) = -\frac{n}{\sigma^2} + \frac{2n\sigma}{\sigma^3} = \frac{n}{\sigma^2}.$$

Recall that we can also work out the Fisher information by taking:

$$\mathcal{I}_{\mathbf{X}}(\sigma) = \text{Var}(s_{\mathbf{X}}(\sigma; \mathbf{X})) = \text{Var}\left(\frac{n}{\sigma^2}(\bar{X} - \mu - \sigma)\right) = \frac{n^2}{\sigma^4} \text{Var}(\bar{X}).$$

The two expressions must be equal, which gives $\text{Var}(\bar{X}) = \sigma^2/n$.

Common errors included failing to apply the factorisation theorem correctly in part (a), overlooking the scalar nature of $\bar{X}$ for minimal sufficiency, and errors in differentiating the log-likelihood in part (b). In part (c), some candidates confused expectations with variances or missed the link between Fisher information and variance, leading to incorrect derivations or conclusions about estimator properties.

Question 3

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with probability density function:

$$f(x; \theta) = \begin{cases} 2\theta^{-2}x & \text{for } 0 \leq x \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

where $\theta > 0$ is an unknown parameter.

- (a) Let $Y = \max_{1 \leq i \leq n} X_i$. Derive the probability density function of Y .

(5 marks)

(b) Find a simple sufficient statistic for θ .

(5 marks)

(c) Given that the maximum likelihood estimator of θ is $\hat{\theta} = Y$, compute:

$$E((\hat{\theta} - \theta)^2).$$

(10 marks)

Reading for this question

Section 3.7.1 of the subject guide explains the derivation of the probability density function of the sample maximum, and Section 2.8.1 covers sufficient statistics, while Section 3.4.1 presents the mean squared error.

Approaching the question

(a) For $x \in (0, \theta)$, we have:

$$F_X(x; \theta) = \int_0^x 2\theta^{-2}t \, dt = \left[\frac{t^2}{\theta^2} \right]_0^x = \left(\frac{x}{\theta} \right)^2.$$

Hence:

$$f_Y(y; \theta) = n(F_X(y; \theta))^{n-1} f_X(y; \theta) = \frac{2ny^{2n-1}}{\theta^{2n}} I(0 \leq y \leq \theta).$$

(b) The likelihood function is:

$$\begin{aligned} L_{\mathbf{X}}(\theta; \mathbf{x}) &= \begin{cases} \frac{2^n}{\theta^{2n}} \prod_{i=1}^n x_i & \text{for } 0 \leq \min_i x_i \leq \max_i x_i \leq \theta \\ 0 & \text{otherwise} \end{cases} \\ &= \frac{1}{\theta^{2n}} I(\max_i x_i \leq \theta) 2^n \prod_{i=1}^n x_i I(\min_i x_i \geq 0). \end{aligned}$$

Hence, by the factorisation theorem, we have that $\max_i X_i = X_{(n)}$ is a sufficient statistic for θ .

(c) The maximum likelihood estimator is $\hat{\theta} = Y = X_{(n)}$. We compute the following quantities. First:

$$E(\hat{\theta}) = \int_0^\theta 2n \frac{x^{2n}}{\theta^{2n}} \, dx = \frac{2n\theta}{2n+1}$$

hence:

$$E(\hat{\theta} - \theta) = -\frac{\theta}{2n+1}.$$

Also:

$$E(\hat{\theta}^2) = \int_0^\theta 2n \frac{x^{2n+1}}{\theta^{2n}} \, dx = \frac{2n\theta^2}{2n+2}$$

hence:

$$\text{Var}(\hat{\theta}) = \frac{2n}{2n+2} \theta^2 - \left(\frac{2n}{2n+1} \right)^2 \theta^2 = \frac{2n\theta^2}{(2n+2)(2n+1)^2}.$$

Therefore:

$$E((\hat{\theta} - \theta)^2) = \text{Var}(\hat{\theta}) + (E(\hat{\theta} - \theta))^2 = \frac{4n+2}{(2n+2)(2n+1)^2} \theta^2.$$

Common errors included incorrect computation of the cdf or pdf of the maximum in part (a), misapplying the factorisation theorem in part (b), and errors in integrating powers of x or algebraic manipulation when calculating expectations and variances in part (c). Some candidates also confused bias with mean squared error or failed to correctly combine variance and squared bias.

Question 4

- (a) State the Neyman–Pearson lemma.

(5 marks)

Reading for this question

Section 5.5.1 of the subject guide presents the Neyman–Pearson lemma.

Approaching the question

Consider a test procedure T of size α for $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$. If the rejection region for T is:

$$R_T = \{\mathbf{x} \in \mathbb{R}^n : L_{\mathbf{X}}(\theta_1; \mathbf{x}) - k_{\alpha} L_{\mathbf{X}}(\theta_0; \mathbf{x}) > 0\}$$

equivalently:

$$R_T = \left\{ \mathbf{x} \in \mathbb{R}^n : \frac{L_{\mathbf{X}}(\theta_1; \mathbf{x})}{L_{\mathbf{X}}(\theta_0; \mathbf{x})} > k_{\alpha} \right\}$$

then T is the most powerful test of size α .

Here $L_{\mathbf{X}}$ is the likelihood function and k_{α} is a constant dependent on α . Note that $\alpha = P_{\theta_0}(\mathbf{X} \in R_T)$.

This was standard bookwork and done well by most candidates.

- (b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be independent exponential random variables, each with probability density function:

$$f_X(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where the rate parameter $\lambda > 0$ is unknown. Suppose we want to test:

$$H_0 : \lambda = \lambda_0 \quad \text{vs.} \quad H_1 : \lambda \neq \lambda_0.$$

- i. Show that we *cannot* find a uniformly most powerful test in this situation.

(7 marks)

- ii. Obtain the likelihood ratio test statistic and also state its asymptotic distribution under H_0 . If:

$$n = 100, \quad \sum_{i=1}^n x_i = 31 \quad \text{and} \quad \lambda_0 = 3$$

would you reject H_0 at the 5% significance level?

Hint: The table below may be useful.

Degrees of freedom, ν	$\chi_{0.10, \nu}^2$	$\chi_{0.05, \nu}^2$	$\chi_{0.025, \nu}^2$	$\chi_{0.01, \nu}^2$
1	2.706	3.841	5.024	6.635
2	4.605	5.991	7.378	9.210
3	6.251	7.815	9.348	11.345
4	7.779	9.488	11.143	13.277
5	9.236	11.070	12.832	15.086

(8 marks)

Reading for this question

Section 5.6 of the subject guide presents likelihood ratio tests.

Approaching the question

- i. Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\lambda; \mathbf{x}) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda n \bar{x}}.$$

First consider testing H_0 against the simple alternative $H_1 : \lambda = \lambda_1$. By the Neyman–Pearson lemma, the most powerful test (MPT) rejects H_0 for large values of the likelihood ratio:

$$LR(\mathbf{X}) = \frac{L_{\mathbf{X}}(\lambda_1; \mathbf{X})}{L_{\mathbf{X}}(\lambda_0; \mathbf{X})} = \frac{\lambda_1^n e^{-\lambda_1 n \bar{X}}}{\lambda_0^n e^{-\lambda_0 n \bar{X}}} = \left(\frac{\lambda_1}{\lambda_0}\right)^n e^{(\lambda_0 - \lambda_1)n \bar{X}}.$$

The issue here is that the form of the MPT depends on the chosen value of λ_1 , as the test statistic is increasing in $\bar{X}$ when $\lambda_1 < \lambda_0$, and decreasing when $\lambda_1 > \lambda_0$. In other words, there is no uniformly most powerful test for H_0 versus the two-sided alternative $H_1 : \lambda \neq \lambda_0$.

- ii. We need the maximum likelihood estimator of λ , which is $\hat{\lambda} = 1/\bar{X}$. The test statistic is, then:

$$LR(\mathbf{X}) = \frac{L_{\mathbf{X}}(\hat{\lambda}; \mathbf{X})}{L_{\mathbf{X}}(\lambda_0; \mathbf{X})} = (\lambda_0 \bar{X})^{-n} e^{n(\lambda_0 \bar{X} - 1)}.$$

We are estimating a single parameter, so the asymptotic distribution of $2 \log LR(\mathbf{X})$ under H_0 is χ_1^2 . For the given values, the observed test statistic value is:

$$2 \log LR(\mathbf{x}) = 2(-100 \times \log(3 \times 0.31) + 100 \times (3 \times 0.31 - 1)) = 0.514.$$

This is lower than the 5% critical value of 3.841, hence we do not reject H_0 .

Common errors included incorrectly concluding that a uniformly most powerful test (UMPT) exists despite the direction of the most powerful test depending on the alternative hypothesis. In part (ii), candidates often miscalculated the likelihood ratio or its logarithm, forgot the asymptotic χ_1^2 distribution, or misinterpreted the critical value comparison when deciding whether to reject H_0 .